

ARM-450 — Purchasing List

- Date: 2026-08-20
- What this covers: everything that must be bought, not printed.
- Printed parts are covered separately in ARM450_DRAWINGS_2D.pdf.

⚠ READ BEFORE YOU SLICE — 2026-08-23

0. THE BUILD MATERIAL IS NOW PLAIN PLA, not carbon-filled — `params.MATERIAL = "PLA"`. Every load case was re-run: 10 of 10 pass, worst-case deflection 0.197 mm against a 0.30 mm limit, and backlash still dominates at 2.25 mm so it is invisible at the tool. Three things change for the better — no hardened nozzle needed (brass is fine), better layer adhesion, and the printed structure is 24 g lighter at 1.24 g/cm³. One thing gets worse: plain PLA shrinks 0.40 % against CF's 0.15 %, which is 0.17 mm on a Ø42 bore — larger than the coupon's whole 0.10 mm bracket. Budget two coupon prints.

1. `output/cad/` contains 12 files that must NOT be printed. Superseded parts (`bearing_yoke`, `wrist_yoke_j5`, `roll_module`, `wrist`, `tool_flange`), models of bought items (`bearing_6806`, `joint_shaft`, `shaft_tube` — the shaft is an aluminium tube), and assembly renders (`arm450_assembly_*`, `arm450_parts_kit`). Print only the 17 files named in the print list below. `python3 final_audit.py` lists them.

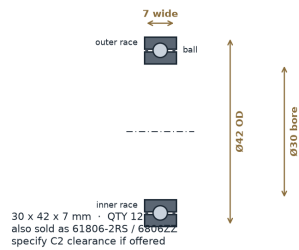
2. The assembled arm is ~1608 g, not 1365 g. The mass gate never had a fastener term: 40 brass inserts, six M5 × 90 preload bolts and ~100 screws come to 244 g, and steel and brass are 6–7× the density of the plastic they hold. That is 158 g over the 1450 g ceiling and it cannot be closed with fastener substitutions alone (everything compatible saves 78 g). This does not affect printing — no printed part changes — but the mass requirement is not met. See `mass_options.py`.

3. The gripper was re-cut on 2026-08-23. Rack teeth were 1.20 mm deep instead of 2.40 (the cutter was centred on the bar face, so half of it cut air), the rack channels broke out of the body by 0.05 mm leaving the jaws in an open slot, and the end tooth land was below the printable floor. Re-export before printing any gripper part.

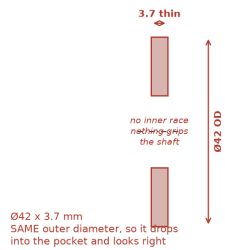
1. The one part that matters most

ARM-450 — parts to buy (drawn to scale from the real dimensions)

BUY THIS
6806-2RS deep-groove ball bearing



NOT THIS
thrust washer (what you already have)



WHY IT MATTERS

A thrust washer takes load along the shaft ONLY. Balls run between two flat plates, so the joint is free to TILT. That tilt is your 28.9 mm wobble.

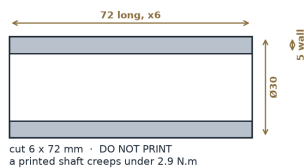
A deep-groove bearing has a grooved raceway that captures the ball, so it resists radial load too. TWO of them, 22 mm apart, resist the tilting moment.

One bearing cannot do it. Two touching cannot do it. It is the SPACING that carries the moment.

28.9 mm -> 0.65 mm

only with the preload screw fitted

Ø30 STEEL TUBE, 5 mm wall



BRASS HEAT-SET INSERTS

knurled, for melting into plastic



M3
Ø4.6 x 5.8
qty 40



M2.5
Ø3.6 x 4.0
qty 12

FASTENERS

item	qty	where
M3 x 12 socket cap	24	link seams
M3 x 16 socket cap	12	collars, wrist
M3 x 20 socket cap	6	bearing PRELOAD
M2.5 x 10	12	tip ears
M2 x 6	24	servo horns
M2.5 x 8	24	servo cases
M4 x 16	4	base to bench
M3 washers	-30	under every head

TWO bearing sizes — read this before ordering

2026-08-21: the wrist changed to a smaller bearing. The arm no longer uses one part number. Ordering 12 x 6806 will leave you unable to build the wrist.

joint	bearing	size	qty
J1 turret, J2 shoulder, J3 elbow	6806-2RS	30 x 42 x 7	6
J4 roll, J5 pitch, J6 tool roll	6706-2RS	30 x 37 x 4	6

Both have the same 30 mm bore, so the aluminium tube, the shaft clamp and the horn adapter are identical at every joint. Only the housing pockets differ — Ø42.00 x 7.00 at J1-J3, Ø37.00 x 4.00 at J4-J6. The two are visibly different sizes, but label the bags: a 6706 dropped into a Ø42 pocket has 2.5 mm of radial slop and will look almost right.

Why the wrist got smaller. The loads there are tiny. Worst case at J4, arm horizontal with a 300 g payload: 3.3 N radial and 390 N·mm of moment, which is a 9.7 N couple across the 40 mm bearing spacing — safety factor 103. A 6806 there was carrying nothing. The swap saves 93 g, about 6 % of the whole arm. Moment stiffness is unaffected because it goes as bearing spacing squared, and the spacing is unchanged at 40 mm.

J1-J3 could take the smaller bearing too on load alone (SF 20-46), and were deliberately left at 6806: those three set tool droop, and a thinner race in a shallower printed pocket is exactly where a printed part loses squareness.

6806-2RS deep-groove ball bearing — quantity 6 (J1, J2, J3)

Size	30 × 42 × 7 mm
Also sold as	61806-2RS (ISO name), 6806ZZ (metal shields — equally fine)
Seals	2RS = rubber both sides; ZZ = metal shields
Clearance	ask for C2 if offered; standard CN is acceptable
Quantity	12 — two per joint × six joints

Why 12 and not 6. A single bearing at one location cannot resist tilting. Two bearings 22 mm apart react the tilting couple across that span. Your link half already has a 7 mm deep Ø42 pocket, so an assembled link takes one bearing in each half.

This is the fix for the 28.9 mm wobble — down to 0.65 mm, and only if the preload screw is fitted as well. Without preload the pair buys 4×, not 44×.

Do NOT reuse the thrust washer you already have

The thrust ball bearing.stl on your Desktop is a Ø42 × 3.7 mm thrust washer.

It shares the same Ø42 outer diameter, so it will drop into the new pocket and look correct. But it is 3.7 mm thick against a 7 mm pocket, has no inner race for the shaft to clamp to, and carries zero moment. It is the part that caused the wobble.

2. Shaft material

item	spec	qty
Aluminium 6061 tube	Ø30 mm OD, 2 mm wall (Ø26 bore)	~500 mm, cut into 6 × 72 mm
M5 × 90 socket cap + nyloc + washers	preload through-bolt	6 sets
Ø3 × 20 spring/roll pin	clamp anti-rotation	12

Why aluminium, and why not steel or printed

There are six joints, so six shafts. That count is what makes the material choice decisive — a per-shaft figure that looks harmless is multiplied by six.

shaft spec	each	× 6	arm total	verdict
Steel Ø30 × 5	222 g	1332 g	2270 g	1070 g over budget
Steel Ø30 × 3	144 g	863 g	1801 g	601 g over
Steel Ø30 × 2	99 g	597 g	1535 g	335 g over
Alu Ø30 × 3	49 g	297 g	1235 g	35 g over
Alu Ø30 × 2	34 g	205 g	1144 g	fits, 56 g spare

Aluminium Ø30 × 2 is the only option that closes the 1.2 kg budget.

Printing the shaft is not ruled out by strength. That was the surprise when the numbers were run: at the 2.94 N·m servo stall a printed Ø30 shaft sees 0.55 MPa of shear (SF 32) and 0.57 MPa of bending (SF 104). Torsion and bending are nowhere near the limit. It fails for two other reasons:

1. The M5 preload thread strips. Threads tapped into a printed part are cut across the layer lines, so they hold at roughly 18 MPa. The preload screw needs 2 N·m, which puts ~16 MPa in the thread — inside the scatter band. Preload is the entire reason the paired bearings work at all (it is what takes joint wobble from 6.55 mm down to 0.65 mm), so a thread that may or may not hold is not acceptable there.
2. The press fit creeps away. A press fit is stored elastic strain. Plastic relaxes under sustained strain, so the 0.02 mm interference bleeds toward zero over days and the inner race starts turning on the shaft.

Both failures are local — one thread, one bearing seat. So the fix is to put metal only where metal is needed:

- The tube provides the two bearing seats and the compression path.
- A steel M5 through-bolt carries the preload, passing right through the Ø26 bore and taking a nyloc on the far side. This is better than the tapped end it replaces: no thread cut in soft aluminium, and the bolt is in pure tension.
- A Ø3 roll pin through the tube replaces the clamp's grub screws and the machined D-flat. It works in double shear (4241 N capacity against the 196 N the joint torque demands, SF 22), so nothing relies on friction and there is no flat to mill.

One thing to check when the tube arrives

Extruded aluminium tube is held to about ± 0.10 – 0.20 mm on OD. A bearing seat wants roughly $+0.011/+0.002$. That is 10–20× looser than needed, so measure the tube with calipers before cutting:

measured OD	what to do
29.98 – 30.01	use as-is, light press
down to 29.92	retaining compound (Loctite 603 or 638) — rated ~15 MPa shear across a gap up to 0.10 mm; this is the standard industrial fix for a slack seat, not a bodge
below 29.92	buy drawn/ground tube, or turn it to size

3. Fasteners

item	qty	where it goes
M3 heat-set inserts, Ø4.6 × 5.8	40	link seams, wrist, collars
M2.5 heat-set inserts, Ø3.6 × 4.0	12	link tip ears

item	qty	where it goes
M3 × 20 socket cap	24	link seams — see the grip note below
M3 × 16 socket cap	12	collars, wrist interfaces
M5 × 90 socket cap + nyloc	6	bearing preload through-bolt — do not skip
M2.5 × 18	12	tip ears — see the grip note below
M2.5 × 8	24	servo case mounting
M2 × 6	24	servo horns
M4 × 16	4	base to bench
M3 washers	~30	under every head
M3 spacer / shim, ~2 mm	6	between the bearing inner races

Seam screw length — get this right or nothing clamps

The seam screw enters from outside, through the flat back face of the tongue half, passes through that half, crosses the seam and threads into a brass insert set into the groove half's boss. So the grip length is the full half-shell depth, not the boss height:

	seam (M3)	tip ear (M2.5)
grip through the tongue half	14.50	14.50
insert length	5.80	4.00
ideal screw	20.30 → M3 × 20	18.50 → M2.5 × 18
usable range	18.0 – 22.0	16.9 – 19.5

Too short and it never reaches the insert. An M3 × 12 — which this list previously specified — stops 2.5 mm short of the seam plane and does not engage anything at all.

Too long and it bottoms in the blind hole and jacks the two halves apart, which recreates the seam gap the tongue-and-groove exists to close. The blind hole is 7.5 deep against a 5.8 insert, so the ceiling is 22.0 mm for M3 and 19.5 mm for M2.5.

The heads sit proud on the outer face — there is no counterbore. That is deliberate: a counterbore in a 2.4 mm wall would leave under 1 mm of material under the head.

The M3 × 20 preload screws are the single most important fastener in the machine. They pull the two inner races of each pair together, removing the clearance that causes both the wobble and the limit-cycle oscillation.

4. Actuators — you already have these

item	qty	note
Waveshare ST3215 serial-bus servo	6	30 kgf·cm at 12 V

All six joints take the same servo. Confirmed against the datasheet: 45.22 × 24.72 × 37.25 mm case, output axis 10.11 mm from one end (not centred), horn bolt circle Ø14.0, encoder 4096 counts = 0.0879°.

5. Consumables

item	note
PLA+CF filament	~600 g for the full set
Hardened steel nozzle, 0.6 mm	CF is abrasive; brass wears out in a few hundred grams
Soldering iron with a heat-set tip	for the brass inserts, 200–220 °C

6. Parts to PRINT — nothing here needs buying

All files are in output/cad/. Thirteen printed pieces, 310 g total, about 356 g of filament once supports and purge are counted — so one 1 kg spool covers the whole arm with room for a reprint.

#	file	part name	qty	g ea	g total
1	link_half_tongue.stl	Link half — TONGUE side	2	47.9	95.9
2	link_half_groove.stl	Link half — GROOVE side	2	44.7	89.5
3	wrist_j4_housing.stl	J4 roll housing	1	40.6	40.6
4	wrist_j5_yoke.stl	J5 pitch yoke	1	34.5	34.5
5	wrist_j6_output.stl	J6 output / tool flange	1	36.9	36.9
6	horn_adapter.stl	Servo horn adapter	6	6.6	39.3
7	shaft_clamp.stl	Shaft clamp (split)	2	6.6	13.3
8	servo_collar.stl	Servo collar	2	22.3	44.7
9	turret_j1.stl	J1 turret	1	73.5	73.5
10	base.stl	Base pedestal	1	135.5	135.5
		TOTAL	19		604

19 printed pieces, 604 g. About 694 g of filament with supports and purge, so one 1 kg spool still covers the whole arm with room for a reprint.

Mass model corrected 2026-08-21. These figures use a 0.90 fill factor, not the 0.55 used previously. The parts are modelled hollow — 2.4 mm link walls, a 3.0 mm base shell — so the STEP volume is already the wall material, and 4 perimeters on a 0.6 nozzle prints a 2.4 mm wall essentially solid. The old 0.55 discounted the hollowing a second time and hid about 240 g.

Arm total 1437 g = 604 printed + 360 servos + 268 bearings + 205 shafts, against an accepted hard ceiling of 1450 g. Headroom is 13 g. The pre-print gate now fails on mass rather than warning, so anything added from here is blocked rather than noted.

horn_adapter is new. It closes the torque path — servo horn to adapter to roll pin to tube to clamp to link. Before it existed there was none: the old solid shaft carried its own horn shoulder, and that went away when the shaft became a bought tube. It reuses the roll-pin hole the tube already has, so the tube needs no extra machining.

Two link halves of each kind make one link; you need two links (upper arm and forearm), which is where the qty 2 comes from. fit_coupon.stl is a test piece and is not part of the arm.

Orientation — decide this before any other setting

Every bearing bore prints with its axis VERTICAL. This is not a preference. Your printer shrinks 0.02 mm in XY and nothing in Z, so a bore drawn by XY nozzle motion comes out uniformly 0.02 mm under — which is the light press fit. Printed on its side, the same bore shrinks in XY but not in Z and comes out oval, which puts back exactly the tilt and wobble the paired-bearing redesign exists to remove.

part	how it sits on the bed
Link halves	Split face flat on the bed, open side up. No supports, and the mating face prints against glass so it is genuinely flat — that flatness is what closes the seam gap you saw on the first print. Bearing bores end up vertical automatically.
Wrist J4 / J5 / J6	Bearing bore axis vertical.
Shaft clamp	Ø30 bore vertical.
Servo collar	Collar axis vertical.
J1 turret	Bearing seat vertical.
Base pedestal	Foot down, upright — the natural orientation, and it puts the J1 seat vertical.

Slicer settings

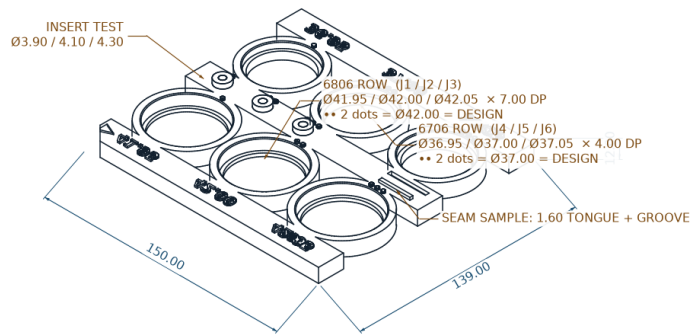
setting	link halves	everything else
Material	PLA+CF	PLA+CF (PLA acceptable for collars/clamps)
Layer height	0.20 mm	0.16 mm
First layer	0.25 mm	0.25 mm
Perimeters	4	6
Top / bottom	5	6
Infill	10-15 % gyroid	100 % clamps & collars, 50-60 % base/turret
Nozzle temp	230-245 °C	230-245 °C
Bed	60-65 °C	60-65 °C
Fan	15 % (0 % first 2 layers)	15 %
Outer perimeter speed	25 mm/s	25 mm/s

Perimeters matter far more than infill here. In bending, resistance goes as $\int y^2 dA$, so the outer quarter of the thickness carries 87.5 % of the moment — and that is exactly where the perimeters live. On a 2 mm wall the walls are the part; raising infill from 20 % to 50 % adds core mass and buys a few percent. Going from 3 walls to 6 converts the whole load-bearing zone from broken infill into continuous aligned extrusions.

Dry the filament — PLA+CF at 50 °C for 6-8 h. Moisture flashes to steam at the nozzle and leaves voids in the weld, attacking the exact property you are printing for.

7. Print this first — the fit-test coupon

FIT-TEST COUPON
PRINT THIS FIRST — it gates every part with a bearing pocket



PART fit_coupon.stl
QTY 1 off
MATERIAL same filament as the parts it gates
MASS 69.6 g · -2.3 h
ORIENT flat, pockets facing UP
ROWS top Ø42 = 6806, J1/J2/J3
bottom Ø37 = 6706, J4/J5/J6
PURPOSE 2.3 h spent here protects the 19 h of parts that carry a bearing pocket
firm thumb press and sits square = right
drops in free = too big
will not start = too small

fit_coupon.stl — 150 × 139 × 12 mm, 70 g, ≈2.3 h.

It carries two rows of bearing pockets stepped 0.05 mm apart — Ø42 for the 6806 and Ø37 for the 6706 — three insert-boss hole sizes, and a tongue-and-groove seam sample. One print tells you which fit is right on this machine before eight hours of parts are committed to it.

The wrist row was added on 2026-09-01. The wrist moved to the 6706 on 2026-08-21 and the coupon was never followed through, so six of the arm's twelve bearings had no fit test at all. Offered up to a Ø42 pocket a 6706 simply drops through — which reads as a faulty bearing when the truth is that the coupon had no station for it.

Every feature is labelled on the part itself. Printed, three pockets 0.05 mm apart are identical to the eye — you press a bearing in, it fits, and you have no idea which one it was, which makes the coupon useless. So it carries:

mark	where	reads
raised numerals 41.95 42.00 42.05	label strip along the top	6806 pocket diameter
raised numerals 36.95 37.00 37.05	label strip along the bottom	6706 pocket diameter
raised 6806 / 6706	right-hand end of each strip	which bearing that row is for
1 / 2 / 3 raised dots	in the ring around each pocket	the same thing, upside-down-proof
1 / 2 / 3 raised dots	below each insert boss	Ø3.9 / Ø4.1 / Ø4.3
raised ► arrow	top-left of the strip	counts always ASCEND left to right

mark	where	reads
clipped corner	top-left	which way up the part reads

Marks stand 0.9 mm proud and are raised, not engraved — in dark carbon-filled PLA an engraved character is nearly invisible, while a raised one catches the light. The numerals are bold at 8 mm so the strokes clear two extrusion widths; anything neater would be dropped by the slicer.

The doubling is deliberate. 05 upside down reads 50 and 95 reads 56, and this is the one part in the build whose entire job is to be unambiguous.

pocket, as modelled	after your 0.02 mm shrink	expected result
Ø41.95	Ø41.93	firm interference press
Ø42.00	Ø41.98	light press — the design value
Ø42.05	Ø42.03	clearance, will wobble

How to read it. Press a 6806 into each pocket by hand. The right one needs firm thumb pressure or a light tap and then sits square on the Ø38 shoulder. If it drops in freely, that pocket is too big; if it will not start at all, too small. Whichever works, set BRG_FIT in cad/params.py to that difference and every part regenerates.

Then test an insert at each of the three hole sizes — the boss must not split — and check that the tongue engages the groove without rocking.

8. Print order

~21 hours of printing on one machine. You have three printers, so the real constraint is not total time but sequence — one 2.3 h test print gates everything else.

stage	part	qty	h ea	h stage	why here
0	fit_coupon.stl	1	2.3	2.3	Gate. Nothing else starts until this is measured.
1	link_half_tongue.stl	2	1.4	2.8	Longest chain of dependent parts — start the moment the gate clears
1	link_half_groove.stl	2	1.3	2.5	Prints alongside the tongue halves on a second machine

stage	part	qty	h ea	h stage	why here
2	wrist_j4_housing.stl	1	1.7	1.7	Wrist is three parts that only mean anything together
2	wrist_j5_yoke.stl	1	1.4	1.4	
2	wrist_j6_output.stl	1	1.5	1.5	
3	shaft_clamp.stl	2	0.3	0.6	Quick; fill gaps between big jobs
3	servo_collar.stl	2	0.9	1.8	
4	turret_j1.stl	1	3.0	3.0	Large but least coupled to the bearing fit
4	base.stl	1	5.6	5.6	Longest single print, least critical — run it overnight
		13		~21 h	

Times assume 0.6 mm nozzle, 4–6 walls, 25 mm/s outer perimeter. Trust your slicer's number over mine — this is a volumetric estimate, not a simulation.

Why this order and not simply largest-first

Stage 0 is a hard gate. The coupon tells you whether Ø42.00 is the right pocket. If it is not, BRG_FIT changes and every part with a bearing pocket is scrapped — that is stages 1, 2 and 4, roughly 19 of the 21 hours. Spending 0.9 h to protect 19 h is the whole point. Do not start stage 1 "to save time while the coupon prints".

Stage 1 before stage 2 because the link halves are the parts you have already printed once and got wrong. They carry the seam, the tongue-and-groove and the bearing bosses — the three things the first print exposed. Getting them in hand early leaves time to react.

Base and turret last because they are the only large parts whose fit does not feed anything else. The base is 5.6 h and would otherwise block a machine all evening for a part that bolts on at the very end.

Running three machines

Put stage 1 on the two machines that can do PLA+CF and keep the third for the small stage-3 parts. That collapses ~21 h into roughly 11 h of wall clock. Print the base

overnight on whichever machine frees up first — it needs no supervision.

If the bearings have not arrived

Print anyway. Plastic is the long-lead item; the bearings drop in afterwards. The one thing not to do is test a joint with the old thrust washers — it will wobble and tell you nothing about this design, because the wobble you measure will be the washer's, not the joint's.

After the parts come off the bed

1. Install heat-set inserts while the parts are still fresh — 200–220 °C, pressed slowly and square.
2. Check the seam closes dry, before any bearing goes in.
3. Press bearings in with the pocket vertical, using a flat plate — never a hammer on the inner race.
4. Assemble one joint completely and check it before building the other five.

8A. End effectors — what to buy and print

Two quick-change tools, one shared bayonet coupling. Neither is on the critical path for the arm; print them after it.

To buy

item	qty	for
SG90 micro servo (9 g)	1	drives the gripper pinion
M3 × 16 thumbscrew	2	the tool lock — one per tool
M3 × 8 socket cap	4	SG90 mounting, dock target bolt-down

The SG90 is the only new actuator in the whole design. The docking tool needs none — its latch is a bayonet and J6 turns it.

To print — 7 parts

order	file	qty	g	note
1	tool_adapter.stl	1	14.4	bolts to J6 once and stays
2	tool_dock.stl	1	48.0	docking probe
2	dock_target.stl	1	12.6	passive port, bench test — not on the arm

order	file	qty	g	note
2	dock_target_sealed.stl	1	12.5	the same port with an O-ring groove, for FLUID transfer
3	tool_gripper.stl	1	43.4	gripper body, SG90 pocket
3	gripper_jaw.stl	2	5.7	100 % infill, print the second mirrored
3	gripper_pinion.stl	1	2.3	100 % infill — small teeth

Same PLA+CF spool as the arm. Print the pinion and the jaw racks at 100 % infill: the teeth are 2 mm and sparse infill leaves them hollow.

Dock first if you want the earliest useful test — dock_target lets the docking interface be proved on a bench with no arm, no fuel and no flight software.

Both docking parts were re-cut on 2026-08-22 and the earlier STLs are dead. The probe exported as four solids (the three latch lugs floated clear of the bore), the target's capture groove had severed its own spigot tip, and the cone was deeper than the spigot was long so the two could not have mated at any depth or angle. Throat is now Ø16, mouth Ø34, cone 9 mm deep, spigot 21 mm. The mate is simulated mesh-on-mesh in check_tool_fit.py — insert, latch, catch, release — see ARM450_END_EFFECTORS.pdf.

Mass — a tool is payload, not arm

The arm is 1365 g against its 1450 g hard ceiling with 85 g spare, and neither tool fits in that. Neither needs to: every load analysis assumed 300 g at the TCP. The heaviest configuration — adapter + gripper + SG90 — is 75 g, a quarter of the allowance. Only one tool is fitted at a time.

One consequence worth knowing — settled

A fitted tool extends the arm past its 450 mm specification. The 450 mm is the arm, base to the J6 tool face; the tool is payload hanging off the end of it.

fitted	reach past J6	max reach	horizontal reach
bare J6 face	—	450 mm	360 mm
pen	30 mm	480 mm	390 mm
gripper	42 mm	492 mm	402 mm
dock probe	36 mm	486 mm	396 mm

Settled 2026-08-22: 492 mm with a gripper is acceptable. 450 mm binds the arm, not the system. Recorded as params.SYSTEM_REACH_MAX = 495 mm and checked in preflight stage 8, so that a later, longer tool cannot quietly push the system past what

was agreed — if one needs more, that is a decision to re-make rather than a number to edit.

The "reach past J6" column is measured off the assembly, not typed in. The dock figure was 43 mm until the capture cone was shortened on 2026-08-22; it is 36 mm now.

9. Where to buy — links

Checked 2026-08-20. My web search only returns US retailers, so the links below are US-based. If you are ordering in India, the two right-hand columns give the exact search term and the local stores that carry this class of part — use those instead. Product links rot; the spec is the thing to match, not the link.

Bearings — 6806-2RS × 6 · 6706-2RS × 6

supplier	note
Bearings Direct — 6806-2RS 30×42×7	~\$14.74 each, 24 h dispatch. Six needed, for J1-J3.
Amazon — 10 pcs 6806-2RS	10-pack, cheapest per unit; order 2 packs.
VXB — 6806-2RS premium sealed	Same-day dispatch.
USA Roller Chain — 6806-2RS	ABEC-3, tighter grade than the Amazon generics.
The Big Bearing Store	Bulk pricing.
123Bearing (EU)	European option.

For the 6706 (30 × 37 × 4), qty 6 — a thin-section deep-groove bearing. Less commonly stocked than the 6806, so order early. Search 6706 2RS 30x37x4, also sold as 61706-2RS. If it proves hard to source locally, the fallback is to run 6806 throughout and accept 1458 g instead of 1365 g — the design works either way, it is only mass.

India: Robu.in bearings category · Robokits India. Search term: 6806 2RS bearing 30x42x7 (also sold as 61806-2RS). If neither stocks it, any local bearing dealer will — this is a standard bicycle bottom-bracket size and is stocked everywhere.

When ordering, get 2RS (rubber sealed), not ZZ (metal shielded). 2RS holds grease better in a joint that mostly sits still and moves slowly.

Aluminium tube — Ø30 OD × 2 mm wall, ~500 mm

supplier	note
Online Metals — 30 mm OD × 2 mm wall × 26 mm ID round tube	Exact match. Alloy is 6060 rather than 6061 — slightly softer, and completely fine here: our safety factors are 21-31, so the alloy difference is irrelevant. Sold in 60" lengths; you need 500 mm.
Chalco Aluminum	True 6061, OD 3-600 mm, wall from 0.5 mm, min order 1 piece on stock sizes.
Online Metals — 6061 pipe range	Browse if the metric size is out of stock.

India: any local aluminium/hardware market stocks this. Search term: aluminium round tube 30mm OD 2mm wall 6061 or 26mm ID. Ask for round tube, not pipe — pipe is sized by nominal bore and you will get the wrong OD.

Measure the OD with calipers before you cut it. See section 2 — extrusion tolerance is 10-20× looser than a bearing seat needs, and this one check decides whether you use it as-is, add retaining compound, or send it back.

Heat-set inserts — M3 × 40, M2.5 × 12

supplier	note
Amazon — 140 pcs M3, 5.7 long × 4.6 OD	Matches our Ø4.6 × 5.8 spec. One pack covers the arm several times.
Prusa — M3 inserts, 100 pcs	Consistent quality, well known dimensions.
MakerTechStore — M3 heat-set	
AndyMark — M3 heat-set	

India: Robu.in and Robokits both stock these. Search term: M3 heat set insert brass knurled 4.6mm.

The critical dimension is OD 4.6 mm, because that is what our Ø4.1 boss hole is sized for. A 5.0 mm OD insert will split the boss. The fit coupon carries Ø3.9 / 4.1 / 4.3 test holes so you can confirm before committing.

Fasteners, roll pins, retaining compound

Standard stock — any fastener supplier, or Amazon/Robu/Robokits. Nothing here is special:

- M5 × 90 socket cap + M5 nyloc + M5 washers, 6 sets — the preload bolts, do not skip
- M3 × 12 / × 16 socket cap, M2.5 × 8
- Ø3 × 20 spring (roll) pins, qty 12
- Loctite 603 or 638 retaining compound — only if the tube measures loose

What you already have

Six ST3215 servos, and the 3 kg-cm-class hardware from the earlier build. Nothing to order there.